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Research Report

Complexity synchronization and targeted restoration in an adaptive predator–prey model

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Abstract

Adaptive systems can achieve similar performance through different internal modes of coordination. We examine complexity synchronization (CS), defined as the correlation between time-dependent scaling exponents, in a reduced adaptive predator–prey model with Selfish Algorithm agents and a Prisoner's Dilemma–like payoff structure. Sliding-window modified diffusion entropy analysis (MDEA) characterizes stripe-crossing event dynamics, whereas detrended fluctuation analysis (DFA) characterizes fluctuation scaling in the original threshold signals. Across the sensing regimes examined, MDEA-based CS increases with cooperative performance, while DFA-based CS exhibits a contrasting association. These estimator-dependent patterns provide complementary descriptions of temporal organization rather than interchangeable measurements of a universal complexity quantity. We also selectively weaken learning in the payoff-sharing subsystem and compare restoration of that known damaged subsystem with a selected non-targeted local intervention and an equal-budget global intervention. Targeted restoration recovers cooperation toward baseline more effectively than these tested alternatives. CS networks describe accompanying changes in scaling dynamics, but the intervention target is specified in advance rather than identified by an independently validated CS-ranking algorithm. The reported correlations and ensemble variability are interpreted descriptively, without inferential claims from nominal correlation p-values. Because the interventions act on learning increments that also affect decisions and payoffs, recovery does not establish a selective causal effect of CS itself. The results support model-specific characterization of coordination and targeted restoration; the use of CS for diagnosis or intervention selection in unknown systems remains a hypothesis requiring independent validation.

Keywords complexity synchronization, adaptive systems, temporal complexity, multi-agent systems, targeted intervention

Significance statement

Teams, organisms and engineered systems can perform equally well while coordinating in very different ways, and standard performance measures cannot tell these apart. We study complexity synchronization: whether the statistical irregularity of interacting components rises and falls together over time. In a simulated predator–prey game, two adaptive agents learn whether to share information, trust each other and share rewards. Two common ways of measuring irregularity give contrasting pictures of how coordination relates to cooperation, so the choice of measure shapes the conclusion. Weakening learning about reward sharing collapses cooperation; restoring that known component recovers it better than equal-cost alternatives. These results describe coordination in one model; using such measures to diagnose failures in real systems remains to be tested.

Introduction

While classical synchronization focuses on the alignment of system states, and criticality emphasizes scaling behavior within individual systems, the present work explores whether coordination in adaptive systems can be understood through synchronization of the evolving complexity of interacting components. The emergence of cooperation in systems composed of self-interested agents is a central problem in the behavioral, biological, and social sciences. The Prisoner's Dilemma (PD) provides a canonical framework for studying this problem by capturing the tension between individual incentives and collective benefits. In repeated interactions, cooperation can arise despite short-term incentives to defect, making the PD a fundamental model for understanding collective organization in adaptive systems (2; 11). More generally, the same framework can be extended to competitive settings, where the temporal organization of interaction may be as important as the average outcome itself.

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In recent years, complexity science has expanded the concept of synchronization beyond the familiar cases of phase locking and amplitude matching (13; 12). This extension is particularly relevant for biological and social systems, where coordination often occurs in signals that are neither periodic nor stationary. Evidence from interpersonal and social neuroscience indicates that coordinated behavior can be accompanied by synchronization across brains, bodies, and autonomic signals during social interaction (4; 6). In such systems, the most informative variable may not be the signal value itself, but the statistical structure of the fluctuations that generate it.

Recent studies (8; 15) proposed that interacting organ networks within the human body exhibit a distinct form of synchronization at the level of temporal complexity. In those studies, cardiovascular, respiratory, and electroencephalographic signals displayed synchronized fluctuations of their scaling indices. This phenomenon was termed *complexity synchronization* (CS). In this framework, the relevant quantities are not the raw signals themselves, but the time-dependent scaling exponents that characterize the evolving statistical structure of their dynamics.

A key implication of this perspective is that coordination may be encoded in how complexity changes over time rather than in static similarity or conventional correlation. Systems can exhibit comparable levels of complexity without interacting, just as they can achieve similar performance through qualitatively different internal organizations. By tracking the coevolution of complexity, CS provides a descriptive summary of how scaling dynamics covary.

One possible mechanism that can generate temporal complexity is self-organized temporal criticality (SOTC), in which intermittent, scale-free events—referred to as Crucial Events—drive continual restructuring of system dynamics over time (10). In such systems, event-driven fluctuations can produce renewal-like statistics and adaptive flexibility. However, this mechanism is not assumed to be universal. In other adaptive systems, coordination may instead be dominated by persistent memory or long-range correlations. We ask how estimator-dependent scaling coordination relates to performance in this model, without assigning a universal temporal mechanism.

To investigate this question, we study adaptive multi-agent systems built from Selfish Algorithm (SA) agents (9), whose decisions are continuously modified by reinforcement-like feedback from prior payoffs within repeated PD-like interactions. Despite purely self-interested behavior, these agents can generate emergent cooperation and collective organization, providing a minimal framework for studying how coordination arises from adaptive interaction dynamics rather than from externally imposed rules.

A central issue in complex systems is that standard performance measures reveal whether the system succeeds, but not how that success is organized internally. A given level of cooperation or payoff may emerge through qualitatively different temporal regimes, including flexible event-driven restructuring, persistence-dominated dynamics, or mixtures of both. The present work describes how different forms of CS covary with performance across the sensing regimes examined.

The diagnostic problem is also an intervention problem. In complex networks, failures may arise from changes in a small subset of components, while restoring function may require identifying a low-dimensional intervention target rather than applying global modification. This idea underlies work on network controllability, realistic control of nonlinear network dynamics, and resilience in complex systems (7; 3; 5). In adaptive systems, however, the relevant intervention targets are often not fixed nodes, but evolving internal variables whose organization changes with learning and interaction. This creates the need for a diagnostic field that links coordination structure to candidate intervention targets.

We examine CS as a descriptive representation that may motivate future diagnostic tests. As summarized in Fig. 1, interactions among adaptive components generate temporal dynamics whose complexity can be quantified by tracking local scaling exponents over time. For simplicity, the MDEA schematic in Fig. 1 shows event extraction using a single threshold. In the actual analyses, however, events are obtained using multiple equally spaced stripes, and each event corresponds to a crossing from one stripe to another. CS is then defined as the correlation between these time-dependent scaling signals across interacting variables. In this framework, CS does not measure performance directly; rather, it characterizes how the internal organization of the system evolves and summarizes covariation among adaptive variables.

To characterize this organization, we use two complementary measures of temporal complexity. Modified diffusion entropy analysis (MDEA) is sensitive to renewal-like, event-driven restructuring, whereas detrended fluctuation analysis (DFA) emphasizes persistent long-range correlations. Together, these methods provide distinct but complementary views of adaptive dynamics and, as illustrated in Fig. 1, define corresponding measures of CS. Comparing MDEA- and DFA-based CS therefore allows us to describe which scaling patterns covary with system performance; predictive intervention selection is not tested here.

In the present paper, we test this framework in a reduced Predator–Prey model and examine how CS relates to cooperation, perturbation, and recovery. In this model, increasing cooperation is associated with stronger MDEA-based CS, consistent with a shift toward more renewal-like, event-driven dynamics. However, this result should be interpreted as model-specific rather than universal. In other adaptive systems, cooperation or performance could instead be associated with synchronized memory-dominated dynamics, in which case DFA-based CS might exhibit a positive relationship with performance. The broader conclusion is therefore not that one scaling framework is always superior, but that comparing complementary forms of CS can characterize temporal coordination; whether it improves intervention selection requires separate validation.

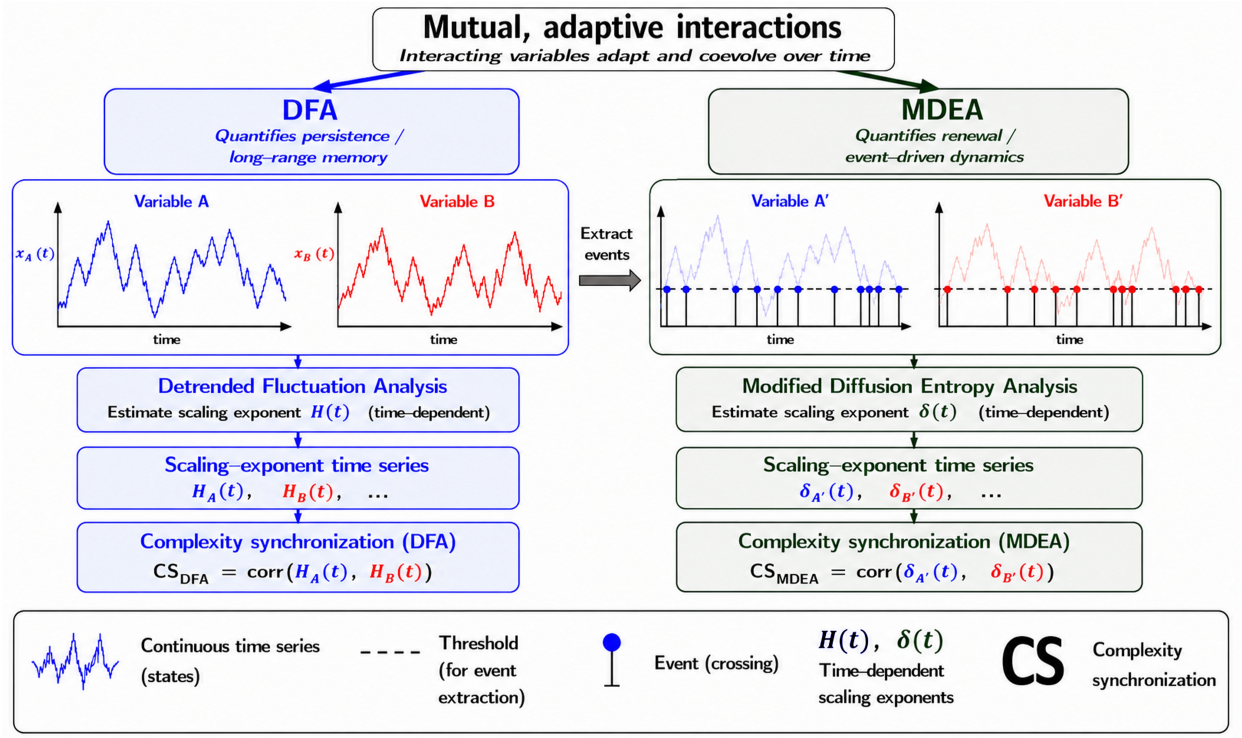

Figure 1 Conceptual pipeline for complexity synchronization (CS). Interacting adaptive variables coevolve over time, local scaling exponents are estimated using complementary analyses, and CS is defined as the correlation between the resulting time-dependent scaling-exponent signals. MDEA emphasizes renewal-like event-driven dynamics, whereas DFA emphasizes persistence and long-range memory.

Methods

Reduced Predator–Prey model

We study a minimal adaptive multi-agent system consisting of two Predator agents and one Prey agent moving on a two-dimensional domain with periodic boundary conditions (Fig. 2). The Predators represent the potentially cooperative elements of the system, while the Prey provides a dynamic and partially unpredictable environment that continuously reshapes the interaction.

Environment and sensing

Each agent interacts with its surroundings through a finite sensing (vision) radius r_S, defining the set of neighboring agents it can detect at each time step. For parameter sweeps, we use the dimensionless sensing-radius ratio $R = r_S/L$, where L is the side length of the square domain; thus $r_S = R \times L$. Based on this local information, agents estimate target directions and update their motion on the periodic domain.

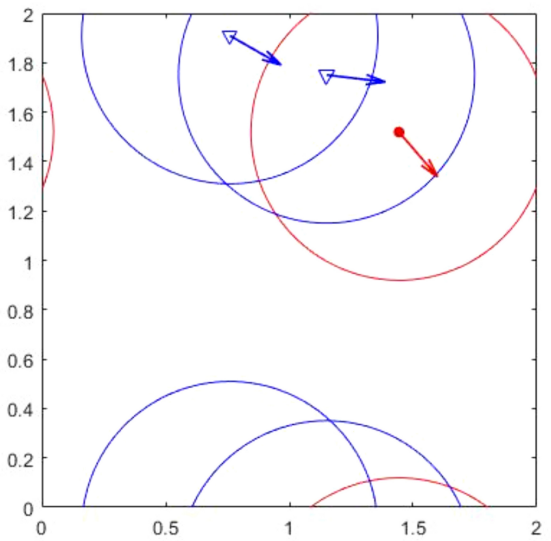

Figure 2 Reduced Predator–Prey environment. Two Predator agents and one Prey move on a two-dimensional periodic domain. Sensing radii define local interaction neighborhoods, and the agents update their headings according to the adaptive decision rules described in the text.

Adaptive decision dynamics

Each Predator is characterized by three internal adaptive variables associated with information sharing, trust, and payoff sharing. These variables encode behavioral propensities and evolve over time through payoff-driven learning. At every time step, they generate probabilistic binary decisions, allowing agents to flexibly switch between alternative actions.

Interaction and information exchange

Predators estimate the Prey's direction by projecting its motion based on local observations. Due to the Prey's deceptive behavior, this estimate is uncertain and can take one of two possible forms. When a Predator relies on its partner, it replaces its estimate with the complementary estimate inferred by the

other Predator. This complementary exchange allows the pair to span the uncertainty in the Prey's trajectory and constitutes the primary coupling mechanism.

The Prey, in turn, anticipates the Predators' motion and selects between two symmetric evasive directions at each step. This stochastic two-option mechanism introduces intrinsic uncertainty and maintains a nonstationary environment.

Payoff and cooperation

Successful capture events—determined by proximity within an interaction radius—can trigger a payoff-sharing interaction between the Predators. In this stage, agents face a Prisoner's Dilemma–like trade-off between cooperation and defection. In the implementation used here, the payoff-sharing stage is entered only if at least one Predator successfully captures the Prey and at least one trust decision is realized. Mutual cooperation is then recorded when both Predators choose to share payoff, i.e.,

$$s_1^{(P)}(t) = s_2^{(P)}(t) = 1.$$

The cooperation rate C_r is defined as the running fraction of these events (i.e., both agents decide to share payoff) over time.

This reduced model combines local sensing, adaptive internal dynamics, complementary information exchange, and a dynamically deceptive environment in a minimal setting that allows the emergence and organization of cooperation to be studied. Full implementation details are provided in the Supplementary Information.

Complexity analysis

The empirical signals analyzed in the present study are the threshold time series

$$I_1,\ T_1,\ P_1,\ I_2,\ T_2,\ P_2,$$

corresponding to information-sharing, trust, and payoff-sharing thresholds for the two Predators. We analyze the temporal complexity of these signals using modified diffusion entropy analysis (MDEA). MDEA converts each threshold trajectory into an event sequence by stripe crossing. Events are defined as crossings of equally spaced amplitude thresholds ("stripes") applied to the signal. Smaller stripe sizes increase the density of detected events and, in the limit, approach noise-dominated regimes, whereas larger stripe sizes capture coarser, more structured fluctuations.

For all reported MDEA analyses, local scaling exponents were estimated using sliding windows of length 10^4 samples with 75% overlap after removal of the first 25% of each simulation as burn-in. Three representative stripe sizes (0.1, 0.01, and 0.001) were examined. Within each window, MDEA was applied to the stripe-crossing event sequence to obtain a local scaling exponent $\delta(t)$. Repeating this procedure across overlapping windows generated the time-dependent scaling signals used for complexity-synchronization analysis.

For comparison, we also estimate scaling exponents $H(t)$ using detrended fluctuation analysis (DFA), which analyzes correlations directly in the original time series rather than through event sequences, and is therefore more sensitive to persistent long-range memory.

CS is then defined as the correlation between the resulting scaling-exponent time series. In this way, CS measures synchronization of changing statistical structure rather than synchronization of raw amplitudes.

Descriptive statistical interpretation

CS values, fitted trends, and ensemble means and standard deviations are reported descriptively. The original plotting scripts used nominal Pearson-correlation p-values for scaling series from overlapping windows and, in some summaries, averages of those p-values across ensembles. Window overlap and temporal dependence invalidate an interpretation based on independent observations, and averaging p-values is not an ensemble significance test. No dependence-aware null analysis or multiple-comparison correction is claimed here. Filled/open markers and solid/dashed network edges are retained solely as legacy plotting styles and are not evidence for significance or absence of an association. Regression R^2 describes fit to the plotted summaries, not out-of-sample prediction or causal evidence.

Renewal-support analysis

To provide supporting mechanistic evidence for the event-driven interpretation of the MDEA results, we also analyzed the renewal properties of stripe-crossing events. These analyses were performed as representative tests on the information-sharing threshold of Predator 1, I_1, rather than on all threshold variables or on the CS pairs directly. The goal was not to define CS, but to examine whether the event sequence extracted from an adaptive threshold exhibits renewal-like organization in the low- and high-cooperation regimes.

For the survival-function aging analysis shown in SI Appendix, Fig. S4, the first half of each simulation was discarded, and a single window of length 10^4 samples was selected from the remaining I_1 trajectory. Stripe-crossing events were extracted using the same stripe sizes used in the MDEA analysis (0.1, 0.01, and 0.001). The inter-event interval τ was defined as the number of samples between successive crossings from one stripe to another. The aging time was fixed at $t_a = 100$, and the aged survival function was compared with a shuffled-aged control obtained by randomly permuting the inter-event intervals before applying the same aging procedure.

For the inter-event-interval autocorrelation analysis shown in SI Appendix, Fig. S5, the first half of the I_1 trajectory was again discarded. Autocorrelations were then computed from the τ sequences extracted from 10 windows of length 10^4 samples with 75% overlap. The plotted curves show the mean autocorrelation across these windows, with variability shown as the window-to-window standard deviation. Because the autocorrelation functions can oscillate around zero, a power-law fit to the upper envelope of the absolute autocorrelation was used as a compact summary of the long-lag decay rate; more negative envelope slopes indicate faster decay of inter-event memory.

Simulation summary

The simulations reported here sweep the dimensionless sensing-radius ratio $R = r_S/L$ over 21 values, use 10 ensembles per condition, and evaluate CS for three representative stripe sizes. Each run contains 10^6 trials, with a burn-in period removed before analysis. The learning increment is fixed at $\Delta_S = 0.1$, and a small stochastic term is included in the threshold update

rule. Full numerical details, ordinary-correlation control analyses, and renewal-related supporting analyses are provided in the Supplementary Information.

Perturbation and rescue protocol

To compare restoration strategies for a predefined perturbation, we applied perturbations to the learning increments of the adaptive thresholds. The threshold order used throughout the analysis is

$$(I_1, T_1, P_1, I_2, T_2, P_2).$$

The baseline learning vector is

$$\mathbf{\Delta}_{\text{base}} = 0.1\,(1, 1, 1, 1, 1, 1),$$

whereas the damaged condition reduces the payoff-sharing learning increments for both agents,

$$\mathbf{\Delta}_{\text{damage}} = (0.1, 0.1, 0.025, 0.1, 0.1, 0.025).$$

Thus, the perturbation selectively weakens the adaptive channel associated with the payoff-sharing thresholds P_1 and P_2, while leaving the information-sharing and trust-related learning increments unchanged.

We compared three rescue strategies with rescue strength $s \in [0, 1]$. Let $b = 0.075s$ denote the increment added to each selected channel. The implemented vectors are

$$\mathbf{\Delta}_{\text{target}}(s) = (0.1, 0.1, 0.025 + b, 0.1, 0.1, 0.025 + b),$$
$$\mathbf{\Delta}_{\text{local}}(s) = (0.1 + b, 0.1, 0.025, 0.1, 0.1 + b, 0.025),$$
$$\mathbf{\Delta}_{\text{global}}(s) = \mathbf{\Delta}_{\text{damage}} + \frac{2b}{6}(1, 1, 1, 1, 1, 1).$$

Thus all strategies add the same total learning-increment budget $2b$. Targeted rescue restores the known damaged P_1–P_2 pair; the selected local control increases the healthy I_1 and T_2 increments above baseline while leaving P_1 and P_2 damaged. This is one specified non-targeted control, not an average over randomly selected targets. Global rescue distributes the budget over all six channels. At $s = 0$ all strategies coincide with damage; at $s = 1$ only targeted rescue exactly restores the baseline vector.

The healthy increments and damaged subsystem are specified in advance. The protocol tests targeted restoration in a known model; it does not establish blinded diagnosis, automatic target selection, or the superiority of CS over other target-selection measurements.

All rescue analyses use 10^6 trials and ENS = 10 independent stochastic ensembles, with distinct random seeds for each ensemble and condition.

A representative segment of the adaptive threshold dynamics used as input to the scaling analyses is shown in Fig. 3.

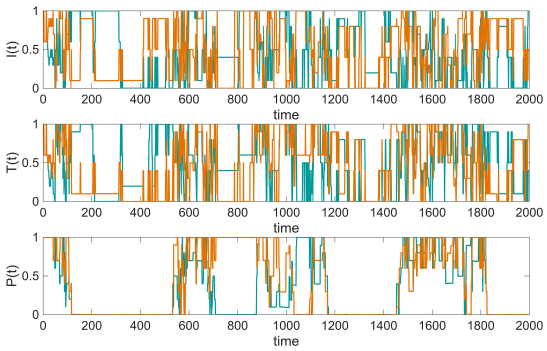

Figure 3 Representative short time series of the three adaptive threshold variables, information sharing $I(t)$, trust $T(t)$, and payoff sharing $P(t)$, over 2000 trials (out of 10^6). The two colored traces correspond to the two Predator agents. These threshold dynamics form the raw signals from which local scaling exponents are extracted using MDEA and DFA.

Results

Learning improves cooperation and payoff

Before turning to CS, we first verify that adaptive learning changes the behavioral outcome of the model. Figure 4 shows that learning increases both the cooperation rate C_r, which is the running fraction of cooperative events per trial, and the payoff rate P_r, which is the running mean payoff per trial of the predators, relative to the non-learning baseline across the dimensionless sensing-radius ratio $R = r_S/L$ (equivalently, $r_S = R \times L$; left panel), and also improves the time-resolved trajectories of these quantities in representative runs (right panel). In both panels, the curves show the average over ten simulations. This establishes that the adaptive threshold dynamics generate a meaningful cooperative regime whose temporal organization can then be analyzed using complexity-based measures.

Scaling estimation from adaptive threshold dynamics

The first step in the complexity analysis is to estimate local scaling exponents from the threshold signals. Figure 5 illustrates this procedure for both MDEA and DFA. The left panels show the resulting time-dependent scaling exponents obtained from sliding windows, whereas the right panels show representative scaling plots from one selected window. These local scaling signals form the basis for the CS measures used throughout the remainder of the Results section.

Cooperation and CS

The main quantitative result of the study is the relationship between cooperative performance and CS. When CS is evaluated using MDEA, the cooperation rate C_r increases with synchronization of scaling dynamics (Fig. 6). In contrast, the same comparison based on DFA yields the opposite trend (Fig. 7).

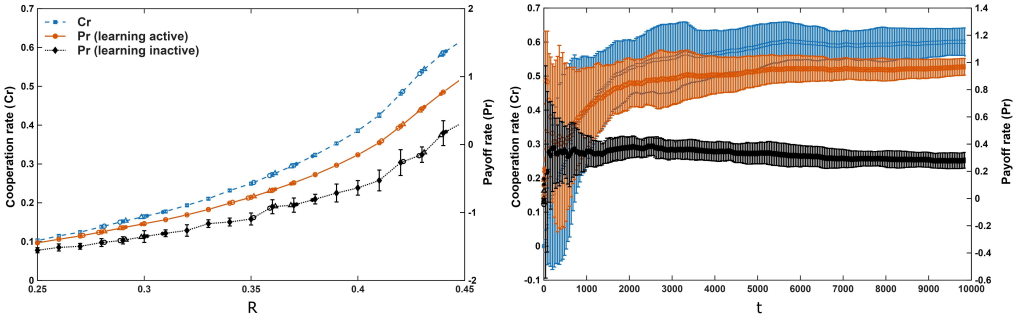

Figure 4 Performance validation of the Predator–Prey model. Left panel: Cooperation rate C_r and payoff rate P_r (at $t = 10^6$) are shown as functions of the sensing radius ratio R for simulations with learning and without learning. Right panel: time evolution of the same quantities for $R = 0.45$.

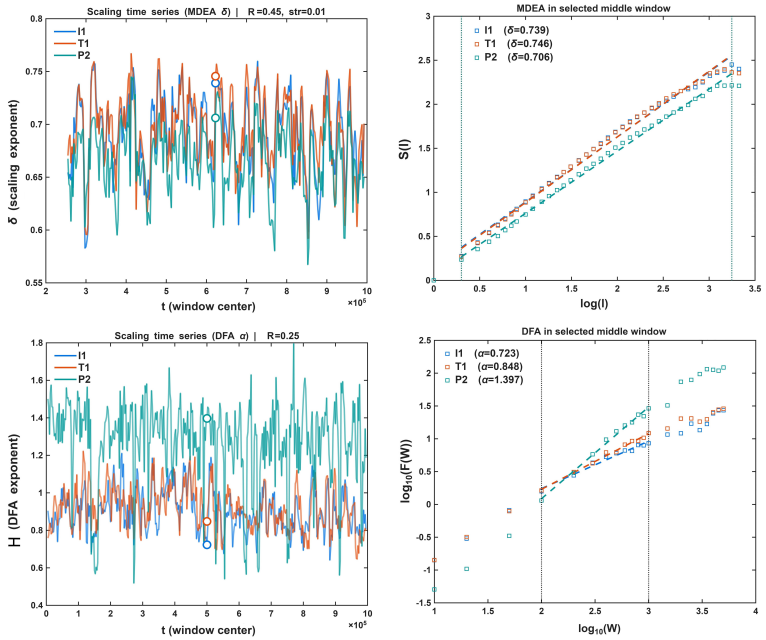

Figure 5 Example of local scaling estimation from the adaptive threshold dynamics. Left panels: scaling-exponent time series obtained from sliding windows using MDEA (top) and DFA (bottom). Right panels: the corresponding scaling plots for a selected window, marked by circles, in the left panels. In the top-right panel, the entropy $S(l)$ is plotted against $\log(l)$, where l is the length of the moving window in the MDEA; in the bottom-right panel, the fluctuation function is plotted on log–log axes, where W is the length of the moving window in the DFA. The fitted slopes define the local scaling exponents used throughout the paper. For the details of MDEA and DFA, see Supplementary Sections S4 and S5, respectively.

Taken together, the plotted results show a positive association between cooperation and MDEA-based CS and a contrasting negative association for DFA-based CS in the conditions examined. These are descriptive associations, not proportionality laws.

The consistency of these trends across multiple threshold pairs, stripe sizes, and interaction regimes shows that the descriptive contrast occurs across several analysis choices within this model; it is not a formal test of a universal relationship. We emphasize, however, that the sign of the DFA-based relation need not be universal across all adaptive systems. In other models, cooperation could in principle be supported by synchronized persistence, stable memory, or common slow environmental forcing, in which case a positive relation between C_r and CS_{DFA} could emerge. The distinctive result here is therefore not that DFA-based CS must always oppose cooperation, but that in the present Predator–Prey model cooperative organization is more strongly aligned with event-driven synchronization than with synchronization of persistent long-range memory. The ordinary Pearson-correlation control analysis is provided in SI Appendix, Fig. S3.

We therefore interpret the contrast between MDEA-based and DFA-based CS not simply as a difference in predictive direction, but as a diagnostic signature of distinct coordination modes. In this sense, the point of the present analysis is not that higher CS must universally imply better performance, but that comparing different forms of CS helps identify how coordination is organized in a given adaptive system.

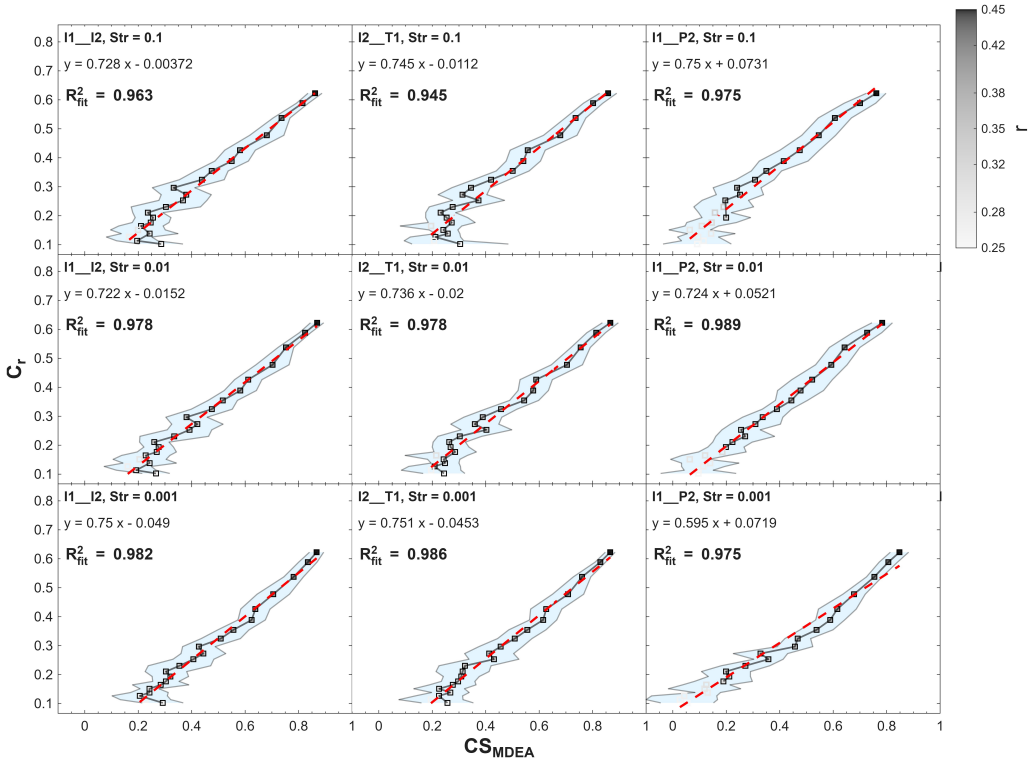

Figure 6 Relationship between cooperation rate C_r and MDEA-based CS. Columns correspond to representative threshold pairs (I_1–I_2, I_2–T_1, and I_1–P_2), and rows correspond to stripe sizes used in MDEA ($\mathrm{Str} = 0.1, 0.01, 0.001$). Each marker represents a simulation at a different sensing ratio $R \in [0.25, 0.45]$, with grayscale indicating the R value (color bar). Filled and open squares retain the original nominal-p-value plotting convention; they are not interpreted as evidence of statistical significance. The plotted correlations and fits are descriptive (see Methods). Red dashed lines show linear fits, with corresponding regression equations and R^2_{fit} values reported in each panel. Shaded bands indicate variability across realizations (mean $\pm$ standard deviation).

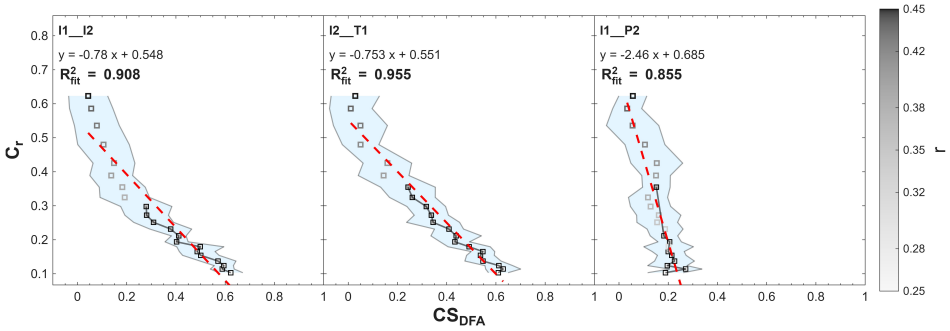

Figure 7 Relationship between cooperation rate C_r and DFA-based CS. Each panel corresponds to a representative threshold pair (I_1–I_2, I_2–T_1, and I_1–P_2), with scaling exponents estimated using detrended fluctuation analysis (DFA). Each marker represents a simulation at a different sensing ratio $R \in [0.25, 0.45]$, with grayscale indicating the R value (color bar). Filled and open squares retain the original nominal-p-value plotting convention; they are not interpreted as evidence of statistical significance. The plotted correlations and fits are descriptive (see Methods). Red dashed lines show linear fits, with corresponding regression equations and R^2_{fit} values reported in each panel. Shaded bands indicate variability across realizations (mean $\pm$ standard deviation).

As a control analysis, we also examined ordinary Pearson correlations computed directly from the raw adaptive threshold signals. SI Appendix, Fig. S3 shows the pair-specific correlations for all 15 threshold pairs as a function of the sensing-radius parameter R, together with the cooperation rate C_r. This analysis separates conventional linear similarity among the raw variables from synchronization of the time-dependent scaling dynamics used to define CS. While some threshold pairs exhibit substantial raw correlations, these correlations are highly pair-specific and summarize a different aspect of the signals; this comparison does not test which representation better selects intervention targets.

Renewal-related supporting analyses

To further interpret the temporal mechanism underlying the observed complexity-synchronization patterns, we examined

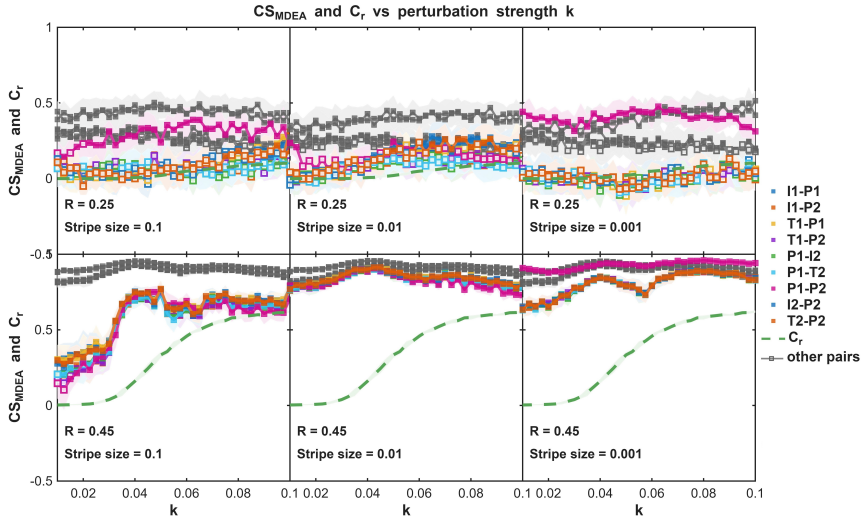

Figure 8 CS, computed using modified diffusion entropy analysis (MDEA), and cooperation rate (C_r) as functions of learning strength k of the SharePay thresholds. The top row corresponds to $R = 0.25$ and the bottom row to $R = 0.45$. Columns represent stripe sizes of 0.1, 0.01, and 0.001, respectively. Colored curves denote CS between selected threshold pairs (I, T, P) involving the payoff sharing thresholds, while gray curves represent other pairwise combinations. The dashed green curve indicates the corresponding C_r. Shaded regions denote variability across ensembles.

stripe-crossing events extracted from the representative information-sharing threshold of Predator 1, I_1. The renewal aging test (SI Appendix, Fig. S4) compares survival functions, aged survival functions, and shuffled-aged controls after discarding the first half of the simulation and analyzing one window of length 10^4 samples. The inter-event autocorrelation analysis (SI Appendix, Fig. S5) uses 10 overlapping windows of length 10^4 samples and quantifies memory in the inter-event interval sequence $\{\tau_i\}$. These analyses are supporting mechanistic evidence rather than additional definitions of CS. They suggest that the high-cooperation regime is accompanied by weaker inter-event memory and a closer approximation to renewal-like event organization, consistent with the positive relationship between MDEA-based CS and cooperation shown in Fig. 6.

CS patterns under perturbation and restoration

The pairwise structure of CS defines a descriptive network over the interacting components of the system. Each pairwise CS value can be interpreted as the strength of alignment between the evolving scaling dynamics of two adaptive variables. In this view, CS is not a single global quantity, but a distributed field that reveals how coordination is organized across information sharing, trust, and payoff sharing.

CS tracks regime-dependent changes in cooperation

We first examined how CS varies with the learning strength k of the payoff-sharing thresholds across different sensing regimes. Figure 8 shows that MDEA-based CS exhibits a clear dependence on k, with a strong contrast between low- and high-cooperation regimes. For $R = 0.25$, CS remains weak across threshold pairs and varies only modestly with k. For $R = 0.45$, CS increases sharply and organizes into a high-synchronization regime across multiple threshold pairs. The corresponding cooperation rate C_r increases with k and reaches substantially higher values in the high-R condition, co-occurring with the emergence of strong event-based CS.

The same analysis performed using DFA-based scaling (Fig. 9) captures changes in coordination, but with weaker resolution in the high-cooperation regime. DFA-based CS remains more variable and less sharply organized than MDEA-based CS, indicating that the event-based scaling measure provides a more resolved description of the cooperative regime in this model. This contrast is consistent with the preceding results showing that cooperation is more strongly aligned with synchronized event-driven restructuring than with synchronized persistence alone.

CS describes changes accompanying perturbation

We next examined changes in CS accompanying a predefined perturbation and its restoration. We damaged the system by reducing the learning increment of the payoff-sharing thresholds P_1 and P_2 from 0.1 to 0.025, while leaving the other adaptive thresholds unchanged. At $R = 0.45$, this perturbation reduced cooperation from the baseline level to a near-zero damaged state (Fig. 10). The CS network also changed, but the change was not a uniform collapse of all links. Instead, the damaged condition retained substantial synchronization while showing weakened links involving the payoff-sharing subsystem. This dissociation indicates that CS and cooperation measure different aspects of the dynamics: CS describes the organization of adaptive temporal structure, whereas C_r measures the behavioral function produced by that organization.

targeted rescue restores cooperation

To examine recovery from the specified perturbation, we compared three rescue strategies using the same damaged system. Non-targeted local rescue applied the intervention budget to a non-targeted pair, equal-budget global rescue distributed the budget across all thresholds, and targeted rescue

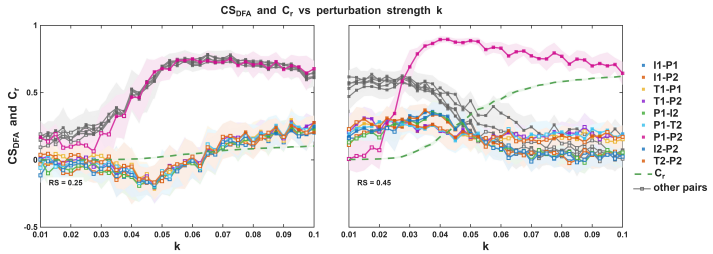

Figure 9 CS, computed using detrended fluctuation analysis (DFA), and cooperation rate (C_r) as functions of learning strength k of the SharePay thresholds. Panel (a) corresponds to $R = 0.25$ and panel (b) to $R = 0.45$. Colored curves represent CS between selected threshold pairs involving the payoff sharing thresholds, while gray curves denote other pairwise combinations. The dashed green curve indicates C_r. Shaded regions denote variability across ensembles.

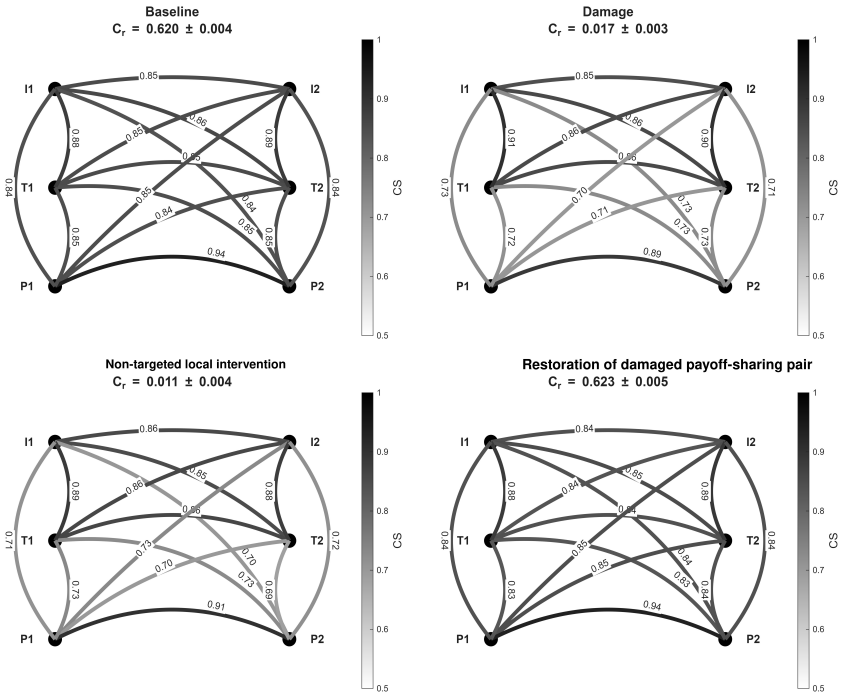

Figure 10 Complexity-synchronization (CS) networks under baseline, perturbation, and rescue conditions (MDEA, $R = 0.45$, ENS $= 10$). Graph representations of pairwise CS among the six adaptive thresholds $(I_1, T_1, P_1, I_2, T_2, P_2)$, computed using modified diffusion entropy analysis (MDEA). Nodes correspond to threshold variables, and edges represent CS links. Edge thickness and grayscale intensity encode CS strength, dashed and solid edges retain the original nominal-p-value plotting convention and do not establish statistical significance, and edge labels report the corresponding CS values. Panels show (top left) baseline, (top right) perturbed (damage), (bottom left) non-targeted local rescue, and (bottom right) targeted rescue. The cooperation rate, C_r (mean $\pm$ SD across ensembles), is reported in each panel. The grayscale color bar spans CS values from 0.5 to 1.

restored the payoff-sharing predefined damaged pair. Figure 11 shows that targeted rescue recovered cooperation monotonically toward the baseline level as rescue strength increased. In contrast, non-targeted local rescue remained close to the damaged level, while equal-budget global rescue produced only partial recovery. Thus, restoration of the specific payoff-sharing subsystem was sufficient to recover cooperative performance, whereas non-specific interventions were inefficient under the same task conditions.

The same restoration protocol was examined in the low-R regime, where cooperation is weaker and DFA-based synchronization provides greater contrast for the relevant coordination structure. The corresponding DFA-based rescue networks and recovery curves are reported in SI Appendix, Figs. S1 and S2. Together, the main and supplementary rescue analyses describe scaling coordination alongside predefined restoration in the two regimes examined, without validating estimator-based diagnosis.

Supporting renewal-related analyses are reported in SI Appendix, Figs. S4 and S5; their interpretation is discussed below as supporting mechanistic evidence rather than as the primary basis of the paper's central claim.

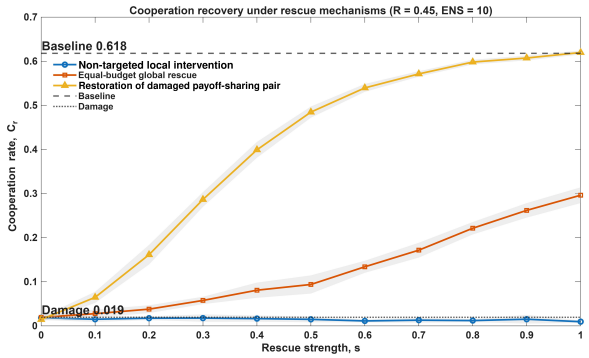

Figure 11 Cooperation recovery under different rescue strategies ($R = 0.45$, ENS $= 10$). Cooperation rate, C_r, is shown as a function of rescue strength, s, for three intervention strategies: non-targeted local rescue, equal-budget global rescue, and targeted intervention. Curves show the mean across ensembles, and shaded bands indicate $\pm$ SD. Horizontal dashed lines denote the baseline and perturbed (damage) levels.

Discussion and limitations

The results describe two estimator-dependent views of temporal coordination in an adaptive predator–prey model. MDEA-based CS increases with cooperation in the plotted sensing conditions, whereas DFA-based CS has a contrasting association. These measures are not interchangeable estimates of a single estimator-independent quantity. The associations, including fitted trends and network patterns, are interpreted descriptively.

Raw Pearson correlation and CS summarize different features of the threshold signals. The raw-correlation control shows strong correlation in the payoff-sharing pair as well as weaker correlations elsewhere. We did not compare interventions selected by raw correlation with interventions selected by CS. Consequently, these results do not establish an incremental diagnostic advantage of CS over ordinary correlation.

The rescue experiments show the functional effect of restoring a specified learning channel. Weakening payoff-sharing learning reduces cooperation while substantial scaling coordination can remain. Restoring the known damaged pair recovers cooperation more effectively than the selected healthy-pair boost and diffuse equal-budget intervention. This comparison is conditional on the model, lesion, and intervention definitions; it does not demonstrate that CS independently identifies the damaged subsystem. No independently computed CS-ranking or search-efficiency experiment is reported.

Learning-increment interventions also directly affect decisions and payoff dynamics. They therefore do not isolate a causal effect of CS itself. Automated diagnosis would require a target-selection rule fixed independently of damage labels, tests on unknown perturbations, comparisons with simpler measurements, and dependence-aware inference. The present results motivate such tests without providing them.

MDEA and DFA emphasize distinct aspects of temporal organization. An estimator that gives a clearer descriptive contrast in one regime need not do so in another. The low-R supplementary restoration analysis uses the same predefined target and therefore illustrates a second regime, not independent validation of a DFA-based diagnostic algorithm.

The renewal-support analyses in SI Appendix, Figs. S4 and S5 should be interpreted in this context. They do not define CS and they are not meant to establish renewal statistics for every threshold pair. Instead, they provide representative mechanistic support, using the I_1 threshold, for the idea that the high-cooperation regime is accompanied by weaker inter-event memory and a tendency toward more renewal-like temporal organization. The aging comparison in SI Appendix, Fig. S4 suggests that temporal ordering of inter-event intervals contributes less additional structure in the high-R regime, while the autocorrelation analysis in SI Appendix, Fig. S5 shows faster decay of inter-event memory.

Interestingly, for the finest stripe size ($\mathrm{Str} = 0.001$) in the high-cooperation regime ($R = 0.45$), the estimated waiting-time exponent approaches $\mu \approx 2$. Using the relation $\beta = 3 - \mu$ for renewal processes with heavy-tailed waiting times (1), this corresponds to $\beta \approx 1$, characteristic of $1/f$-like temporal organization. Although this observation is based on a representative renewal analysis rather than an ensemble estimate, it is nevertheless consistent with the tendency toward scale-free event dynamics in the high-cooperation regime.

These observations help explain why an event-based complexity measure such as MDEA provides a clearer diagnostic of cooperative organization in the high-interaction regime, whereas DFA remains useful for regimes in which persistence-dominated dynamics are more prominent.

This interpretation also clarifies the distinction between CS and complexity matching. In complexity matching, information transfer is favored when interacting systems have similar complexity levels (1; 14). In CS, by contrast, the changes in complexity become synchronized under interaction (8; 15). The present results support the latter description and suggest that synchronized changes in complexity may inform candidate intervention targets as well as characterize coordination.

Potential applications to biological, social, and engineered systems remain hypotheses. Before CS is used for intervention selection, its added value relative to existing measurements and its robustness across estimators, perturbations, and systems must be established.

Conclusion

This study provides a descriptive analysis of complexity synchronization in an adaptive predator–prey model. MDEA- and DFA-based scaling series exhibit contrasting associations with cooperation. A separate perturbation experiment shows that restoring the predefined damaged payoff-sharing learning subsystem recovers cooperative performance more effectively than the tested non-targeted local and equal-budget global alternatives. These results characterize coordination alongside model-specific restoration; they do not establish automatic diagnosis, CS-based search efficiency, statistical significance from the original nominal p-values, or a general causal/control principle.

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Author contributions

K.M. conceived the study, developed the model and analyses, and drafted the manuscript. All authors contributed to interpretation of the results, provided critical feedback, revised the manuscript, and approved the final version.

Competing interests

The authors declare no competing interests.

Data availability

The MATLAB codes and their corresponding .mat data files are publicly available at `https://github.com/Korosh137/Complexity-synchronization-as-a-diagnostic-and-control-principle`.

Supplementary material

Supplementary material is provided in the accompanying SI Appendix.

References

1. Gerardo Aquino, Mauro Bologna, Bruce J. West, and Paolo Grigolini. Transmission of information between complex systems: 1/f resonance. *Physical Review E*, 83(5):051130, 2011.
2. Robert Axelrod and William D. Hamilton. The evolution of cooperation. *Science*, 211(4489):1390–1396, 1981.
3. Sean P. Cornelius, William L. Kath, and Adilson E. Motter. Realistic control of network dynamics. *Nature Communications*, 4:1942, 2013.
4. Guillaume Dumas, Jacqueline Nadel, Robert Soussignan, Jacques Martinerie, and Line Garnero. Inter-brain synchronization during social interaction. *PLOS ONE*, 5(8):e12166, 2010.
5. Jianxi Gao, Baruch Barzel, and Albert-László Barabási. Universal resilience patterns in complex networks. *Nature*, 530(7590):307–312, 2016.
6. Ivana Konvalinka, Dimitris Xygalatas, Joseph Bulbulia, Uffe Schjødt, Else-Marie Jegindø, Sebastian Wallot, Guy Van Orden, and Andreas Roepstorff. Synchronized arousal between performers and related spectators in a fire-walking ritual. *Proceedings of the National Academy of Sciences*, 108(20):8514–8519, 2011.
7. Yang-Yu Liu, Jean-Jacques Slotine, and Albert-László Barabási. Controllability of complex networks. *Nature*, 473(7346):167–173, 2011.
8. Korosh Mahmoodi, Scott E. Kerick, Paolo Grigolini, Piotr J. Franaszczuk, and Bruce J. West. Complexity synchronization: a measure of interaction between the brain, heart and lungs. *Scientific Reports*, 13(1):11433, 2023.
9. Korosh Mahmoodi, Bruce J. West, and Cleotilde Gonzalez. Selfish algorithm and emergence of collective intelligence. *Journal of Complex Networks*, 8(3):cnaa019, 2020.
10. Korosh Mahmoodi, Bruce J. West, and Paolo Grigolini. Self-organizing complex networks: individual versus global rules. *Frontiers in Physiology*, 8:478, 2017.
11. Martin A. Nowak. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006.
12. Arkady Pikovsky, Michael Rosenblum, and Jürgen Kurths. *Synchronization: A Universal Concept in Nonlinear Sciences*. Cambridge University Press, 2001.
13. Steven H. Strogatz. *Sync: How Order Emerges from Chaos in the Universe, Nature, and Daily Life*. Hyperion, 2003.
14. Bruce J. West, Elvis L. Geneston, and Paolo Grigolini. Maximum information exchange between complex networks. *Physics Reports*, 468:1–99, 2008.
15. Bruce J. West, Paolo Grigolini, Scott E. Kerick, Piotr J. Franaszczuk, and Korosh Mahmoodi. Complexity synchronization of organ networks. *Entropy*, 25(10):1393, 2023.